# Week 7 & 8 Assignment – Lottery Numbers

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**DSC 640**

## Purpose

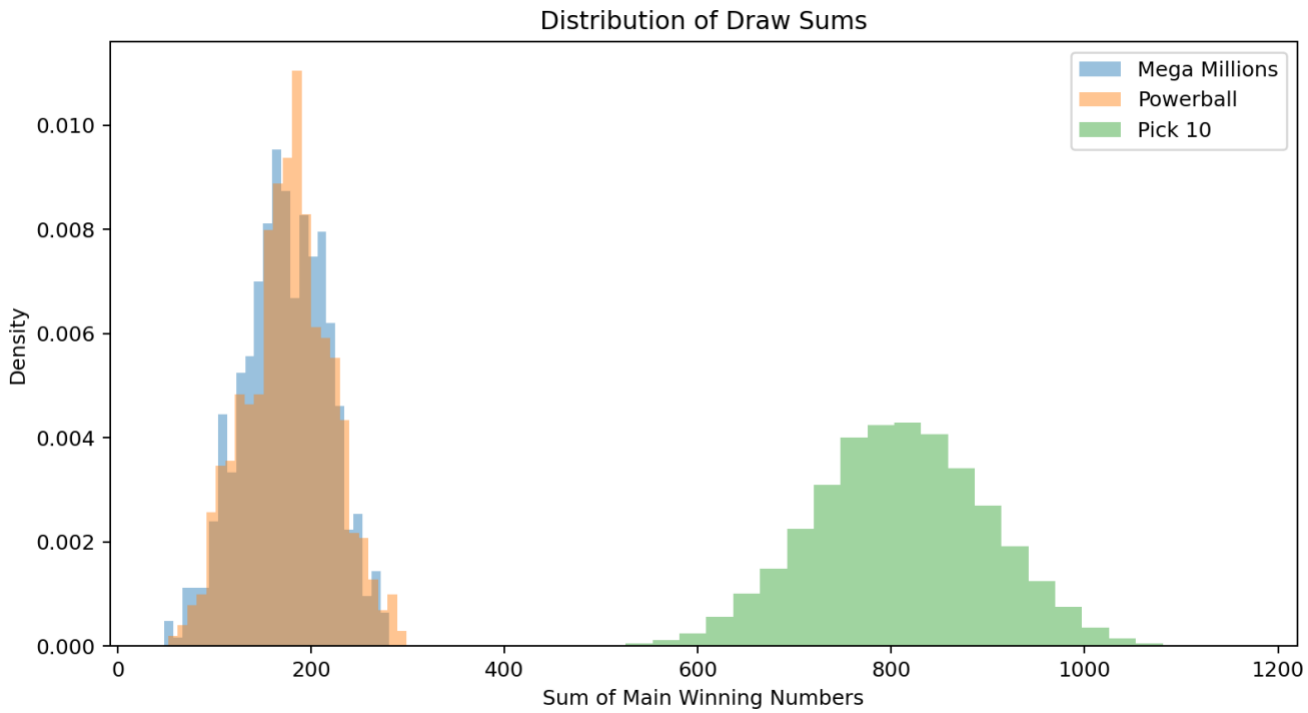
Call to Action: For this assignment, the results of the lottery tend to look like a pattern but using the three datasets from this assignment do not provide evidence that the drawings are systematically biased. We want to look at the lottery as a form of entertainment and have play responsibly. We never want to select numbers because they might appear as due.

## Key Findings

| Game | Draws Analyzed | Current Main Range | Chi-Square p-value | Interpretation |
|---|---|---|---|---|
| Mega Millions | 675 | 1–70 | 0.597 | Consistent with uniform frequency |
| Powerball | 1026 | 1–69 | 0.396 | Consistent with uniform frequency |
| Pick 10 | 13570 | 1–80 | 0.938 | Consistent with uniform frequency |

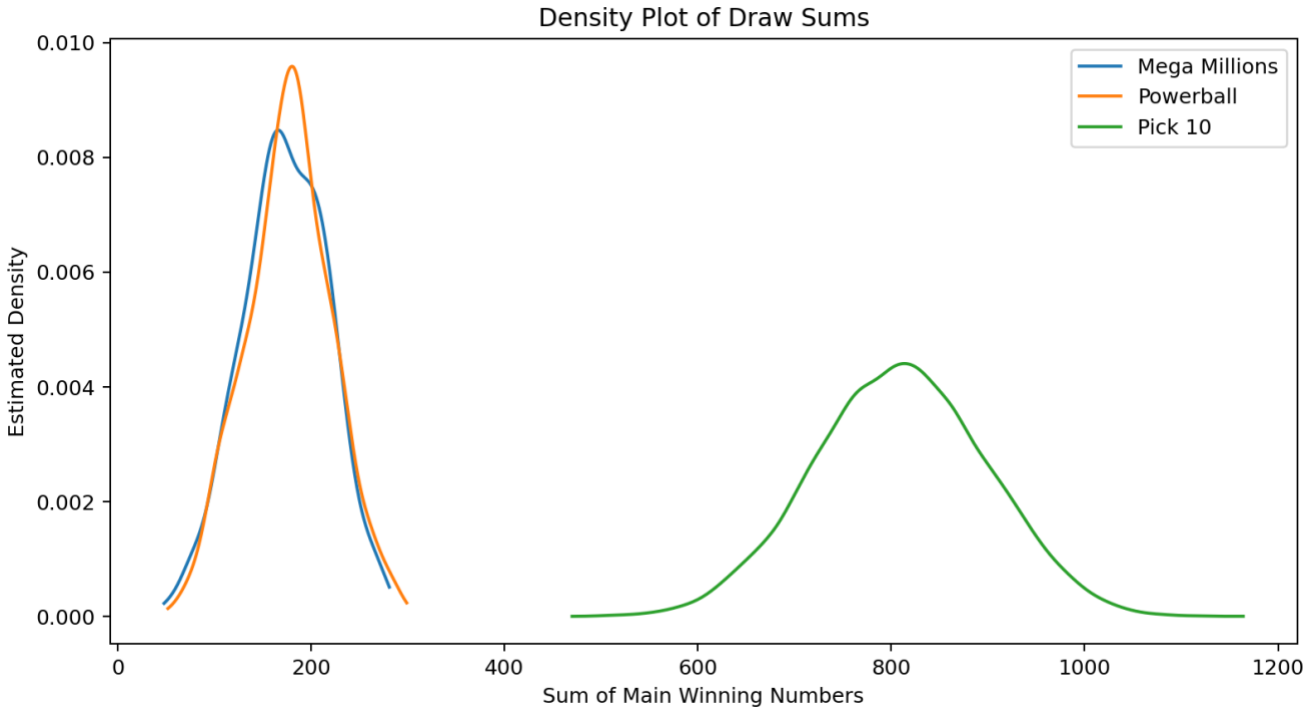
As you can see, the p-values are all well above 0.05 which means that the differences between the common and uncommon numbers are not statistically unusual under a random drawing model. This doesn't prove that the randomness is perfect, but it does show us that the datasets do not justify a claim that the jackpot system is rigged.

## Visual 1 — Histogram: Draw-Sum Distributions



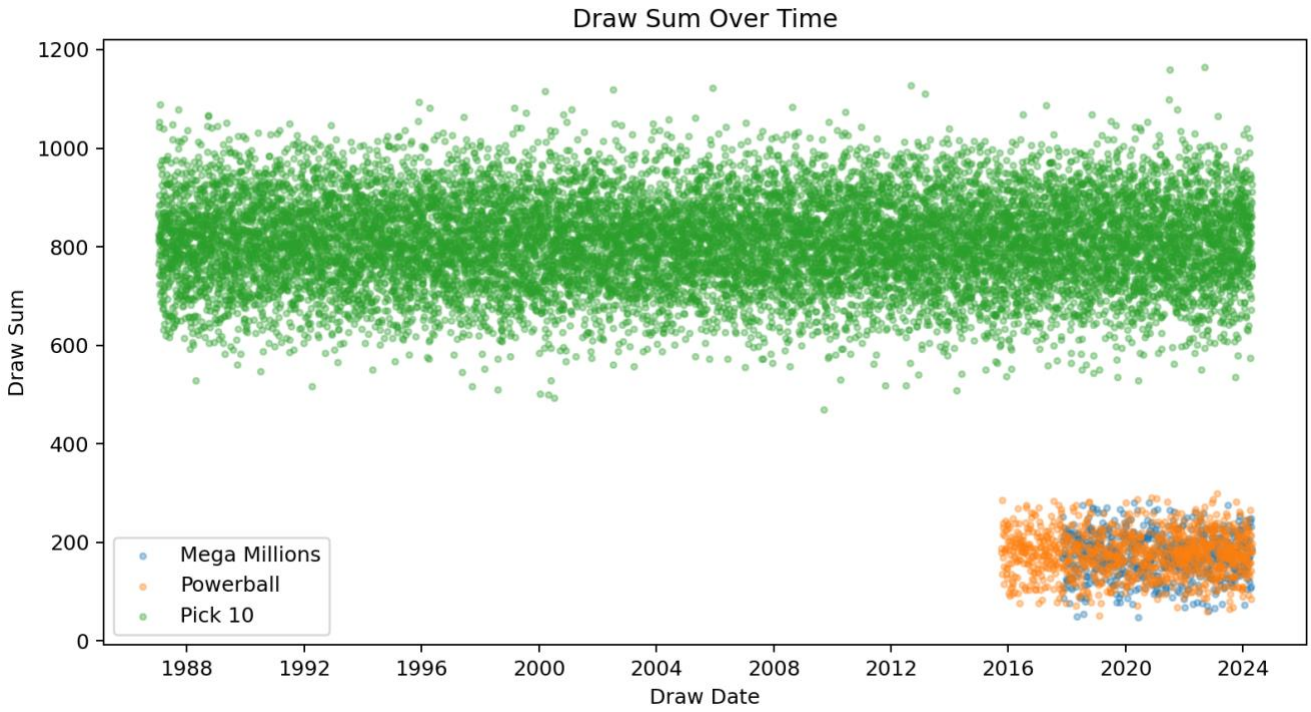

Most draw sums cluster near the center of each game's possible range. Extreme low and high sums occur less often, which is expected from random combinations.

## Visual 2 — Density Plot: Smoothed Draw-Sum Patterns



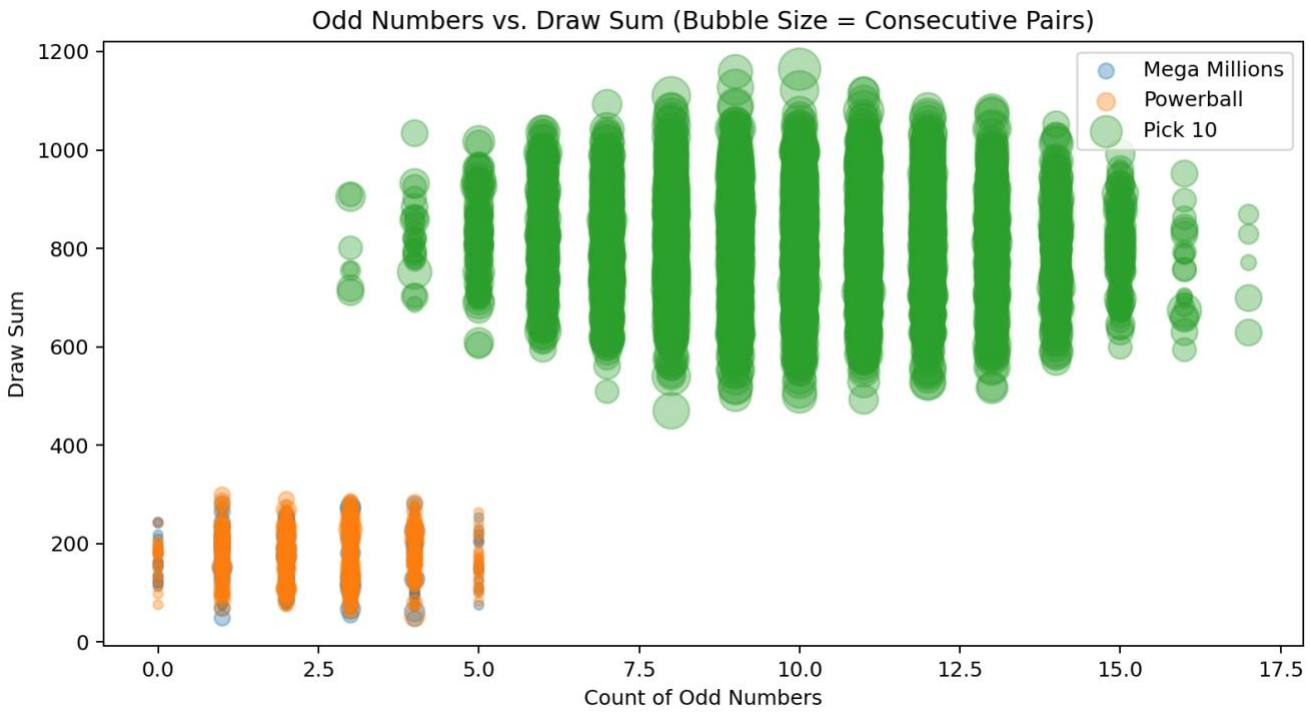
The smooth curves reinforce that each game has a stable central distribution rather than a hidden upward or downward pattern.

## Visual 3 — Scatterplot: Draw Sums Over Time



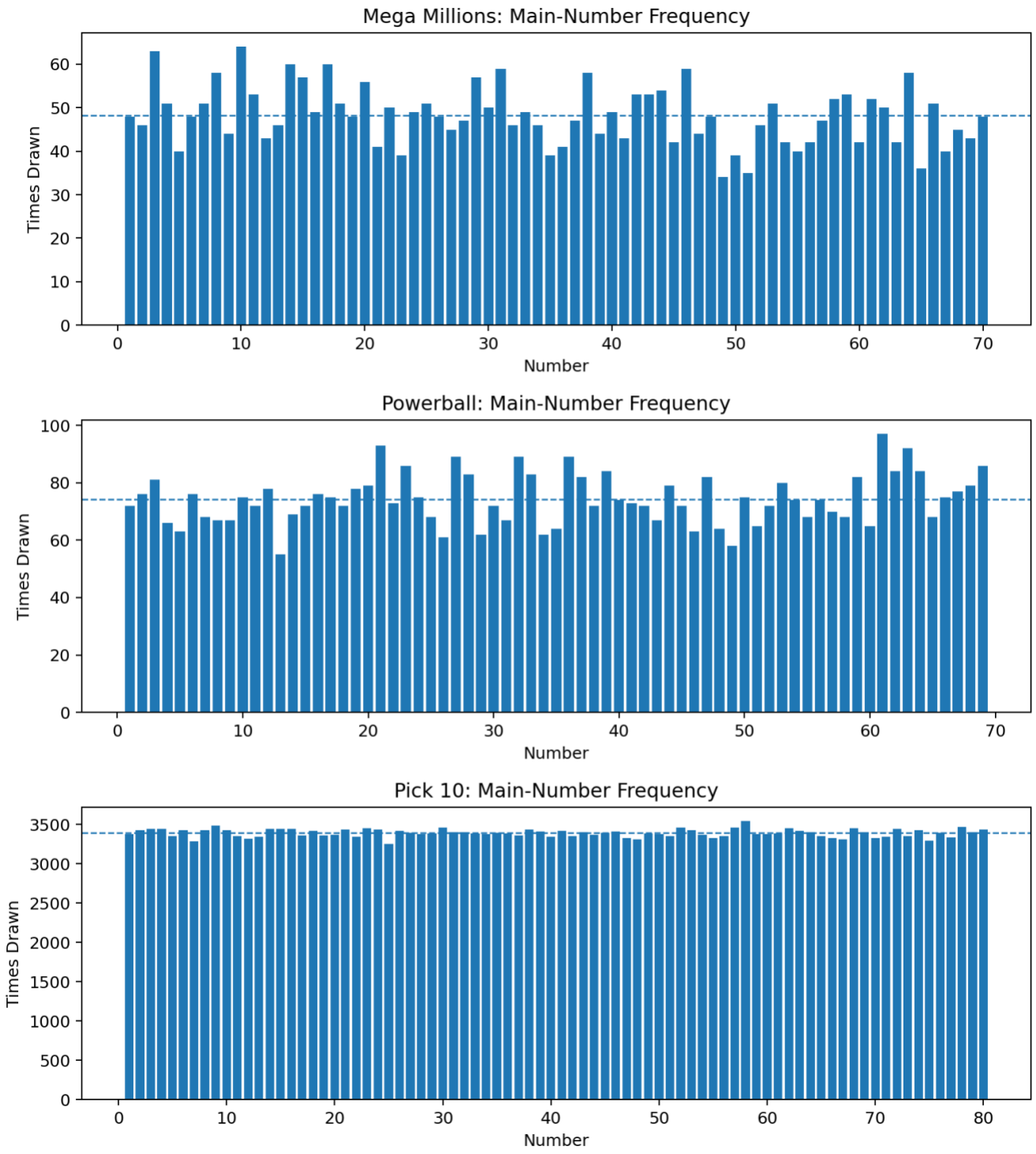

Draw sums fluctuate irregularly over time. No persistent trend suggests that winning numbers are becoming systematically larger or smaller.

# Visual 4 — Bubble Chart: Odd Counts, Sums, and Consecutive Pairs



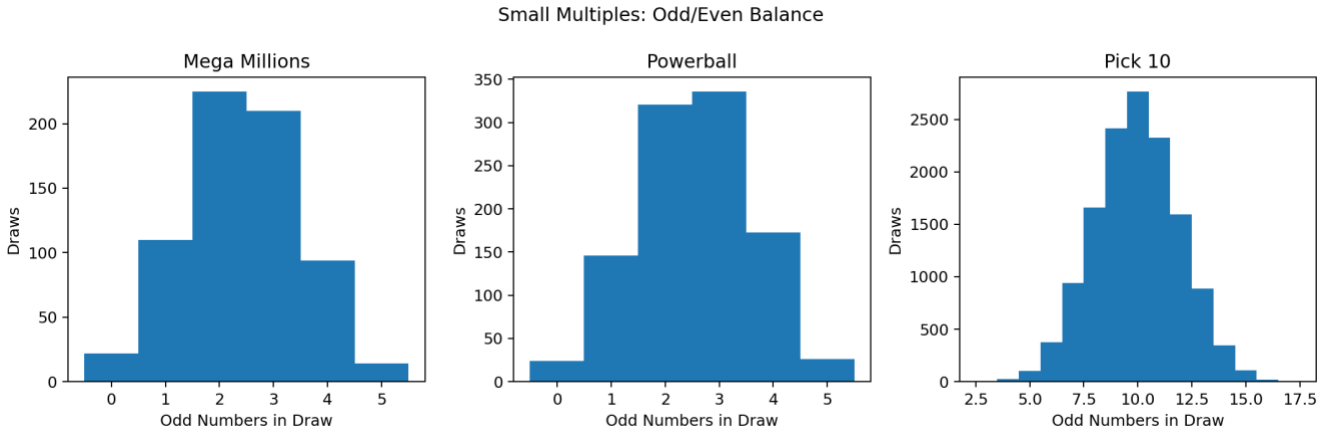

Combinations that look unusual—many odd numbers or consecutive pairs—still occur naturally. Bubble size shows the number of adjacent-number pairs.

## Visual 5 — Small Multiples: Number Frequencies




Every game has hot and cold numbers, but the differences remain within the range expected from random variation.

## Visual 6 — Small Multiples: Odd/Even Balance




Odd and even values generally balance around the center. Pick 10 concentrates near ten odd numbers because twenty numbers are selected per draw.

## 250–500 Word Design and Ethics Summary

Our intended audience for this assignment is the general public, especially those who play the lottery who might believe that frequently drawn numbers are more likely to win again or that some of the rare numbers are due to be drawn. We are assuming that the audience doesn't have advanced statistical knowledge, I wanted to ensure we use plain language, familiar comparisons, and visual patterns and minimized including technical formulas. The goal of this assignment is to persuade our audience rather than informing them as the audience should take away that the random data naturally creates patterns such as streaks, clusters, or hot numbers. The call to action for this assignment is to treat the lottery as a form of entertainment and to play responsibility while also avoiding changing your behavior based on misleading number patterns.

I included a word document as a medium for this assignment, while also including six visuals created in Power BI, which are included above. We used Gestalt principles when designing this assignment through proximity, similarity, and common alignment. We grouped related statistics together and used consistent labeling throughout each of our charts to ensure the audience could easily compare. I also used colors to compare and contrast the games without implying that one game is going to be a safer or better option. I included a few other design points such as headings that were larger than the supporting text, white space, and chart titles.

For ethical considerations, duplicates and blank records were checked before handling, dates were converted to valid date fields, and the winning number strings were split into individual numbers. Mega Millions and Powerball changed their number matrices over time, so the frequency comparison uses only the current-format periods. I wanted to highlight any filtering in order to avoid mixing formats. We used the data from the state of New York and the New York State Gaming Commissions. Some of the risks we face include overstating randomness, implying future outcomes, or encouraging gambling. We minimized these risks as much as possible by avoiding giving recommendations and stating that historical records cannot predict the next drawing.

# Power BI Build Specifications

**Histogram:** Axis: DrawSum bins; Values: count of distinct DrawDate; Legend: Game.

**Density Plot:** Use a Python visual in Power BI with DrawSum and Game; use scipy.stats.gaussian_kde.

**Scatterplot:** X-axis: DrawDate; Y-axis: DrawSum; Legend: Game; Details: DrawDate.

**Bubble Chart:** X-axis: OddCount; Y-axis: DrawSum; Size: ConsecutivePairs + 1; Legend: Game.

**Small Multiples Frequency:** X-axis: Number; Y-axis: count of Number; Small multiples: Game.

**Small Multiples Odd Count:** X-axis: OddCount; Y-axis: distinct count of DrawDate; Small multiples: Game.